



MATHEMATICS (EXTENSION 2)

2012 HSC Course Assessment Task 1

November 28, 2011

General instructions

- Working time – 55 minutes.
- Commence each new question on a new page.
Write on both sides of the paper.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

- ☐ 12M4A – Mr Weiss
- ☐ 12M4B – Mr Ireland
- ☐ 12M4C – Mr Fletcher

NAME: # BOOKLETS USED:

Marker's use only.

QUESTION	1	2	3	Total	%
MARKS	$\overline{20}$	$\overline{20}$	$\overline{18}$	$\overline{58}$	

Question 1 (20 Marks)		Commence a NEW page.	Marks
(a)	i. Find the values of the real numbers a and b such that $(a + ib)^2 = 5 - 12i$.		2
	ii. Hence or otherwise, solve the equation $ix^2 + 3x + (3 - i) = 0$.		2
(b)	Evaluate $(1 + i)^9$.		2
(c)	On an Argand diagram, sketch the locus of a point which satisfies		
	i. $ z - i = 4$.		2
	ii. $(z - 1)(\bar{z} + 1) = 3$.		2
	iii. $ z + 2 = z - 2 $.		2
	iv. $\text{Arg}\left(\frac{z + 3i}{z - 1}\right) = \frac{\pi}{6}$.		2
(d)	i. If $z_1 = 3 + 4i$ and $ z_2 = 13$, find the greatest value of $ z_1 + z_2 $.		3
	ii. If $ z_1 + z_2 $ has its greatest value and also $0 < \arg z_2 < \frac{\pi}{2}$, express z_2 in the form $a + ib$ where a and b are real.		3

Question 2 (20 Marks)

Commence a NEW page.

Marks

- (a) If ω represents one of the complex cube roots of unity, evaluate **3**

$$(1 - \omega^8)(1 - \omega^4)(1 - \omega^2)(1 - \omega)$$

- (b) In an Argand diagram, the points P , Q and R represent the complex numbers z_1 , z_2 and $z_2 + i(z_2 - z_1)$ respectively.

i. Show that $\triangle PQR$ is right-angled. **2**

ii. Find in terms of z_1 and z_2 the complex number represented by the point S such that $PQRS$ is a rectangle. **2**

- (c) i. Use De Moivre's Theorem to show that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. **2**

ii. Deduce that $8x^3 - 6x - 1 = 0$ has solutions $x = \cos\theta$, where $\cos 3\theta = \frac{1}{2}$. **3**

iii. Find the roots of $8x^3 - 6x - 1 = 0$ in the form $\cos\theta$. **2**

iv. Hence evaluate $\cos\frac{\pi}{9}\cos\frac{2\pi}{9}\cos\frac{4\pi}{9}$. **2**

- (d) For the complex numbers

$$z_1 = (1 + i)^2 \quad \text{and} \quad z_2 = \sqrt{2} \left[\cos\left(-\frac{5\pi}{6}\right) + i \sin\left(-\frac{5\pi}{6}\right) \right]$$

i. Express z_1 in modulus-argument form. **1**

ii. Express z_2 , $\overline{z_2}$ and iz_1 in Cartesian form. **3**

Question 3 (18 Marks)		Commence a NEW page.	Marks
(a)	i. Use the remainder theorem to find a factor of $x(x-1) - a(a-1)$.		1
	ii. By division or otherwise, find the other factor.		2
(b)	Prove that if the polynomial $P(x)$ has a root of multiplicity m then $P'(x)$ has a root of multiplicity $(m-1)$.		3
(c)	Given $(x-2)^2$ is a factor of $x^3 - 3x^2 + ax + b$, find the value of a and b and hence factorise the polynomial over $\mathbb{C}$.		4
(d)	The equation $x^3 + 3x^2 + 4x - 7 = 0$ has roots α , β and γ . Find the equation with roots 2α , 2β and 2γ .		3
(e)	If $(1+i)$ is a root of the equation $x^3 - ax^2 + bx - 4 = 0$ where $a, b \in \mathbb{R}$, find the values of a and b and hence solve the equation.		5

End of paper.

Q1

$$(a) \quad (i) \quad (a+ib)^2 = 5-12i$$

$$\therefore \begin{cases} a^2 - b^2 = 5 \\ 2ab = -12 \end{cases}$$

$$\therefore a^2 - \left(\frac{6}{a}\right)^2 = 5$$

$$a^4 - 5a^2 - 36 = 0$$

$$(a^2 - 9)(a^2 + 4) = 0$$

$$\therefore a = \pm 3 \quad (a \text{ is real})$$

$$\therefore a = 3, b = -2 \quad \text{or} \quad a = -3, b = 2.$$

$$\left[\text{ALT.} \quad \begin{aligned} a^2 - b^2 &= 5 \\ ab &= -6 \end{aligned} \right]$$

$$\therefore \text{by inspection, } a = 3, b = -2 \quad \text{or} \quad a = -3, b = 2.$$

(Note: answer can also be written as $\pm(3-2i)$.
But "a = ± 3 , b = ± 2 " loses a mark.)

$$(ii) \quad ix^2 + 3x + (3-i) = 0$$

$$\therefore x = \frac{-3 \pm \sqrt{9 - 4i(3-i)}}{2i}$$

$$= \frac{-3 \pm \sqrt{5-12i}}{2i}$$

$$= \frac{-3 \pm (3-2i)}{2i}$$

$$= \frac{-2i}{2i} \quad \text{or} \quad \frac{-6+2i}{2i}$$

$$= -1 \quad \text{or} \quad 3i+1$$

(* answers must be simplified for full marks!)
i.e. no i's on denominator

Q1 - continued

$$(b) \quad 1+i = \sqrt{2} \left(\frac{1}{\sqrt{2}} + i \cdot \frac{1}{\sqrt{2}} \right) = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$$

$$\therefore (1+i)^9 = (\sqrt{2})^9 \operatorname{cis} \frac{9\pi}{4}$$

$$= 16\sqrt{2} \operatorname{cis} \frac{\pi}{4}$$

$$= 16(1+i) = 16 + 16i$$

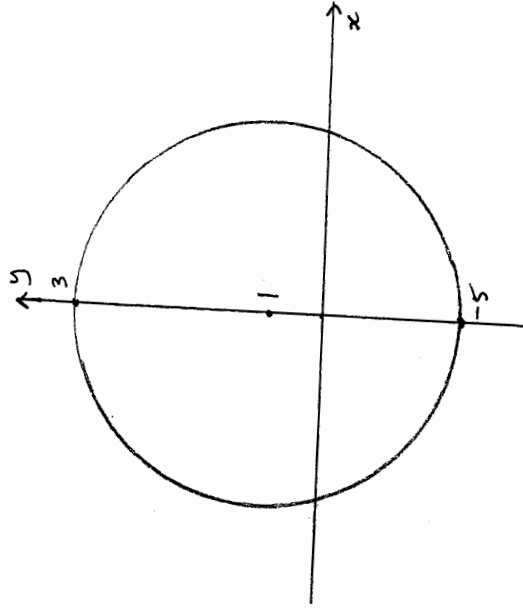
$$\left[\text{ALT.} \quad (1+i)^2 = 2i \right]$$

$$\therefore (1+i)^8 = (2i)^4 = 16$$

$$\therefore (1+i)^9 = 16(1+i) = 16 + 16i$$

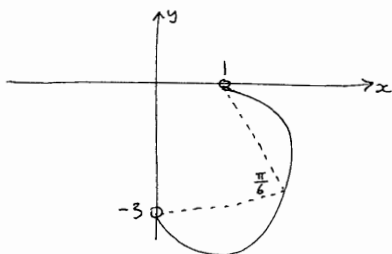
$$(c) \quad (i) \quad |z-i| = 4 \quad \therefore |z-(0+i)| = 4$$

This is a circle, centre (0,1), radius 4



Q1 - continued

(iv) $\text{Arg} \left(\frac{z+3i}{z-1} \right) = \frac{\pi}{6}$



(* Locus is a major arc, in Q4, and end points are open circles at (0,-3) & (1,0).)

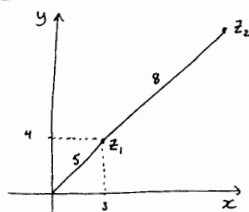
(d) (i) $|z_1 + z_2| \leq |z_1| + |z_2|$ (Δ inequality)

$$= \sqrt{3^2 + 4^2} + 13$$

$$= 18$$

i.e. max. value is 18

(ii) $|z_1 + z_2|$ has its max. when z_1 & z_2 are collinear.

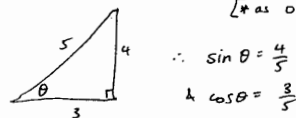


Thus $z_2 = k z_1$
 $= k(3+4i)$
 $\therefore (3k)^2 + (4k)^2 = 13^2$
 $\therefore k = \frac{13}{5}$

$\therefore z_2 = \frac{13}{5} (3+4i)$
 $= \frac{39}{5} + \frac{52}{5}i$
 (alt. $\rightarrow$)

Q1 - continued.

(d) (ii) ALT. $\therefore \arg z_2 = \arg z_1 = \tan^{-1} \left(\frac{4}{3} \right) = \theta$ ($\checkmark$)
 [as $0 < \arg z_2 < \frac{\pi}{2}$]



$\therefore \sin \theta = \frac{4}{5}$
 $\& \cos \theta = \frac{3}{5}$ ($\checkmark$)

$\therefore z_2 = 13(\cos \theta + i \sin \theta)$
 $= \frac{39}{5} + \frac{52}{5}i$, as before. ($\checkmark$)

20

Q1 - continued

(c) (ii) $(z-1)(\bar{z}+1) = 3$

$\therefore z\bar{z} - \bar{z} + z - 1 = 3$

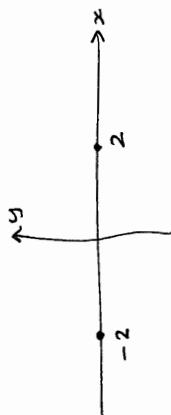
$\therefore x^2 + y^2 - (x-iy) + (x+iy) - 1 = 3$

$\therefore x^2 + y^2 + 2yi = 4$

Equating Re and Im parts,

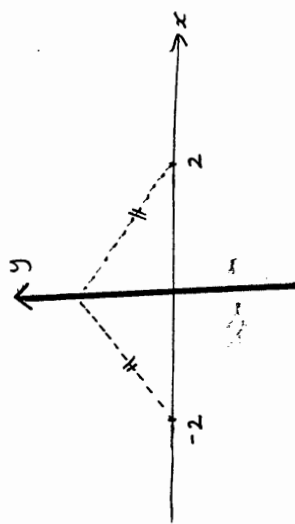
$\therefore x^2 + y^2 = 4$ and $y = 0$

Thus the locus comprises 2 points:



(iii) $|z+2| = |z-2|$

The locus is all those points equally distant from (-2,0) and (2,0)
 i.e. the y-axis ($x=0$).



(* axes are difficult to show when a locus, So best to write the equation $x=0$.)

Q2) (a) $1 + \omega + \omega^2 = 0$
 $1 = -\omega - \omega^2$
 $\omega^3 = 1$

$$\begin{aligned} (1-\omega^3)(1-\omega^2)(1-\omega^2)(1-\omega) &= (1-(\omega^3)^2\omega^2)(1-\omega^4)(1-\omega^2)(1-\omega) \\ &= (1-\omega^2)(1-\omega)(1-\omega^2)(1-\omega) \\ &= \{(1-\omega^2)(1-\omega)\}^2 \\ &= (1-\omega-\omega^2+\omega^3)^2 \\ &= (1+1+1)^2 \\ &= 9 \end{aligned}$$

(c) i/ $(\cos \theta + i \sin \theta)^3 = \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta$

also $(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta$

$\therefore \cos 3\theta = \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta)$
 $= 4 \cos^3 \theta - 3 \cos \theta$

ii/ Now $8 \cos^3 \theta - 6 \cos \theta - 1 = 0$
 $\therefore 4 \cos^3 \theta - 3 \cos \theta = \frac{1}{2}$

$\therefore \cos 3\theta = \frac{1}{2}$

$\therefore 3\theta = \frac{\pi}{3} \pm 2n\pi$

iii/ $\theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}, \dots$

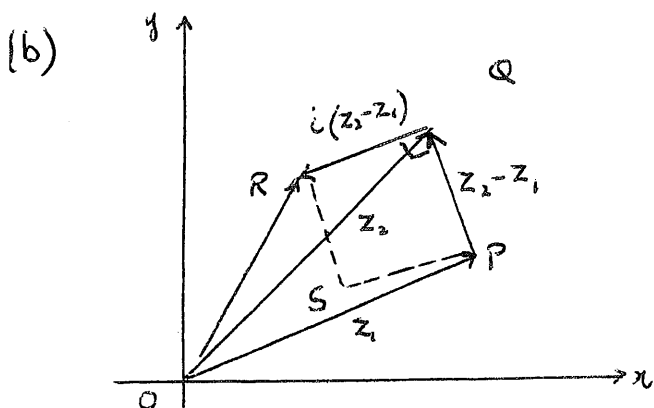
$\therefore$ roots of $8x^3 - 6x + 1 = 0$

are $\cos \frac{\pi}{9}, \cos \frac{5\pi}{9}, \cos \frac{7\pi}{9}$

$\cos \frac{5\pi}{9} = -\cos \frac{4\pi}{9}$

iv/ $\cos \frac{7\pi}{9} = -\cos \frac{2\pi}{9}$

$\therefore \cos \frac{\pi}{9}, \cos \frac{2\pi}{9}, \cos \frac{4\pi}{9} = \frac{1}{8}$



i) $\angle PQR = \frac{\pi}{2}$

(d)

$\therefore \triangle PQR$ is right angled at Q

ii) If PQRS is a rectangle

$\vec{PS} \parallel \vec{QR}$

and hence $\vec{PS}$ also represents $i(z_2 - z_1)$

Now $\vec{OS}$ is the vector sum of $\vec{OP}$ and $\vec{PS}$

Hence S represents $z_1 + i(z_2 - z_1)$

(d) i/ $z_1 = (1+i)^2$
 $= 2i$
 $= 2 \cos \frac{\pi}{2}$

ii/ $z_2 = \sqrt{2} \left(-\frac{\sqrt{3}}{2} - \frac{i}{2} \right)$
 $= -\frac{\sqrt{6}}{2} - \frac{\sqrt{2}i}{2}$

$\bar{z}_2 = -\frac{\sqrt{6}}{2} + \frac{\sqrt{2}i}{2}$

$i z_1 = i \times 2i$
 $= -2$

Q3) (a) i/ $f(x) = x(x-1) - a(a-1)$ (e)
 $f(a) = a(a-1) - a(a-1)$
 $= 0$

$\therefore (x-a)$ is a factor

ii/

$$\begin{array}{r} x + (a-1) \\ x-a \overline{) x^2 - x - a^2 + a} \\ \underline{x^2 - ax} \\ ax - x - a^2 + a \end{array}$$

$\therefore f(x) = (x-a)(x+a-1)$

(c) $P(x) = x^3 - 3x^2 + ax + b$

$P'(x) = 3x^2 - 6x + a$

$P'(2) = 0$ because $(x-2)$ is a double factor

$0 = 12 - 12 + a \Rightarrow a = 0$

$P(2) = 0$

$0 = 8 - 12 + 0 + b \Rightarrow b = 4$

$\therefore a = 0 \quad b = 4$

$P(x) = (x-2)^2(x+1)$

(d) If $x = 2\alpha, 2\beta, 2\gamma$

then $\frac{x}{2} = \alpha, \beta, \gamma$

$x^3 + 3x^2 + 4x - 7 = 0$

thus

$\left(\frac{x}{2}\right)^3 + 3\left(\frac{x}{2}\right)^2 + 4\left(\frac{x}{2}\right) - 7 = 0$

$\frac{x^3}{8} + \frac{3x^2}{4} + 2x - 7 = 0$

$x^3 + 6x^2 + 16x - 56 = 0$

If $1+i$ is a root so is $1-i$

Let the other root be α

then $1+i + 1-i + \alpha = a$

$\alpha + 2 = a$

$(1+i)(1-i)\alpha = 4$

$2\alpha = 4$

$\alpha = 2$

$\therefore a = 4$

$\therefore$ roots are $1+i, 1-i, 2$

$(1+i)(1-i) + 2(1+i) + 2(1-i) = b$
 $b = 6$

(b) Let the root of multiplicity m be α

$P(x) = (x-\alpha)^m Q(x) \quad Q(\alpha) \neq 0$

$\therefore P'(x) = m(x-\alpha)^{m-1} Q(x) + (x-\alpha)^m Q'(x)$
 $= (x-\alpha)^{m-1} [m Q(x) + (x-\alpha) Q'(x)]$
 $= (x-\alpha)^{m-1} S(x)$

$\therefore P'(x)$ has a root of mult. $(m-1)$